



Prediction analysis of vortex-induced vibration of long-span suspension bridge based on monitoring data

Shiqiao Xu^a, Rujin Ma^a, Dalei Wang^{a,*}, Airong Chen^a, Hao Tian^b

^a College of Civil Engineering, Tongji University, Shanghai, China

^b Zhejiang Institute of Communications, Zhejiang, China

ARTICLE INFO

Keywords:

Long-span suspension bridge
Monitoring data
Vortex-induced vibration
Wind field data
Vibration acceleration data
Prediction model

ABSTRACT

Vortex-induced vibration (VIV) is a kind of wind-induced vibration that occurs at low wind speeds. The actual effect of VIV in bridges is different from that in wind tunnels and has a severe impact on traffic safety. Therefore, to guide the bridge management department in taking reasonable control measures in the initial stage of VIV and ensure that vehicles pass over the bridge safely, an identification model for VIV is established based on a large amount of data collected from Zhoushan Sea Crossing Bridge. Firstly, the wind field and vibration characteristic parameters of VIV are calculated based on data measured from 2014 to 2016 by anemometers and acceleration sensors. Then, by the big-data significant differences method and distribution function fitting analysis, two wind field parameters and two vibration response parameters are determined to serve as indexes in the identification of VIV. Finally, based on a parameter optimization algorithm, the VIV identification model is established and the data of 2017 are used to test the prediction model. The results show that the recognition rate of VIV is 89% and the identification model can be used to warn about the occurrence of VIV in practical engineering.

1. Introduction

In the wake of the introduction of new computing theories, materials, and construction technology, the structure of long-span bridges is softer and their crossing ability has shown a continuous increase. Therefore, their stiffness and resistance gradually decrease (Ehsan and Scanlan, 1990; Li et al., 2011). However, owing to the increasing streamlining of the main beam section, the wind sensitivity of bridges also increases (Mashnad and Jones, 2014). Consequently, the problem of wind-induced vibrations is becoming increasingly prominent, and vortex-induced vibration (VIV) is one of the most common types. Although VIV is not a divergent vibration and occurs under relatively low wind speed, the vibration amplitude is large enough to affect the safety and comfort of the drivers. The Tacoma suspension bridge in the United States crashed with a strong wind-induced vibration under the action of a 19 m/s wind speed in July 1940. Based on the experience and lessons of this accident, the influence of wind-induced vibration has been fully taken into consideration during subsequent bridge design (Yoshimura and Mizuta, 2001; Laima et al., 2013). Accidents like that of the Tacoma Bridge will not happen again, but studies on the vortex-shedding characteristics of the bridge girder section and basic theories of VIV are still relatively defective. Therefore, VIV of long-span bridges is still very common worldwide.

In February 1995, the steel bridge beam of the Tokyo Bay Bridge in Japan had an obvious first-order vertical vibration mode with a maximum amplitude of 50 cm; the wind speed was 17 m/s and the wind direction was nearly perpendicular to the axis of the bridge (Zuo et al., 2008; Yang et al., 2015). Under the action of a storm, the Rio-Niteroi Bridge in Brazil had a strong vertical VIV, which was followed by several vertical VIV events with wind speed from 15.3 m/s to 16.7 m/s in 1980 (Marra et al., 2015). The Second Severn Crossing Bridge in Great Britain incurred an obvious VIV event in the first year after it was opened to traffic (MacDonald et al., 2002). Moreover, a large deviation was found between the measured value and the recommended value of the bridge standard.

More attention is being paid to studying VIV by bridge wind engineers because VIV occurs frequently in long-span bridges. Nowadays, the main methods used to study VIV are wind tunnel tests, computational fluid dynamics (CFD) numerical simulations, and field measurements (Yu et al., 2019). Although wind tunnel testing is the main method used to study wind-induced vibrations in long-span bridges, experimental results have obvious size effects due to the limitation of the wind tunnel size (Hua et al., 2015). As a new method for the study of wind-induced vibration, CFD numerical simulation has been widely used in bridge wind engineering. However, owing to the influence of calculation errors and turbulence model errors, its accuracy and reliability are not high. In

* Corresponding author. 1239 Siping Road, Shanghai, China.

E-mail address: wangdalei@tongji.edu.cn (D. Wang).

<https://doi.org/10.1016/j.jweia.2019.06.016>

Received 22 February 2019; Received in revised form 30 May 2019; Accepted 23 June 2019

Available online 3 July 2019

0167-6105/© 2019 Elsevier Ltd. All rights reserved.

addition, many grids are required to obtain reliable results, which makes the computation cycle very long (Zhou et al., 2015). Anemometers, accelerometers, displacement meters, data acquisition systems, and other instruments are essential tools for the field-measurement method (Cantero et al., 2018). Comprehensive wind field characteristics and wind-induced vibration response data can be obtained directly. Therefore, the field-measurement method is considered to be the most suitable method for the study of wind-induced vibration and can describe the phenomenon of wind-induced vibration on a bridge structure in an actual wind environment.

Some scholars have carried out short-term field measurements to study the wind field of some long-span bridges. Bietry et al. (1995) studied the relationship between the wind field characteristics and structural modal parameters in detail on the basis of the Saint-Nazaire cable-stayed bridge in France, including the wind speed, wind direction, horizontal wind, vertical fluctuating wind turbulence intensity, turbulence integral scale, and power spectral density function. Frandsen (2001) synchronously measured the wind pressure on the box girder and the vertical acceleration of the main beam of the east bridge on the Danish sea. The measured results showed that the vertical VIV mostly occurs in the main beam when the wind speed is small, the turbulence is low, and the wind direction is basically perpendicular to the axis of the bridge. Kubo (2004) mounted three accelerometers and anemometers on the Second Severn Crossing cable-stayed bridge in Britain. It was found that the turbulence ratio of the pulsating wind was lower than the design value, but the power spectrum of pulsating wind was significantly higher than that of the design spectrum. Marra et al. (2017) arranged five anemometers along the bridge axis in the main span of the Akashi Kaikyo Bridge in Japan and analyzed the wind field characteristics of two typhoons passing by the bridge. The correlation analysis between the wind speed and the displacement showed that the ratio of maximum displacement to static displacement was less than the Japanese specification design value (1.55) when the wind direction was perpendicular to the bridge axis. Li et al. (2017) applied an automatic and fast clustering algorithm to recognize VIVs on the bridge deck using long-term-monitoring datasets. The acceleration amplitude and the frequency ratio were selected as indicators for the clustering of VIVs from other vibrations.

A series of investigations about the prediction and prevention of VIV have been conducted by scholars. In addition, some progress has been made in the analysis of VIV by the field-measurement of some long-span bridges. However, owing to the complexity and uncontrollability of the field environment, the variability of short-term test results is large. In addition, the existing analysis of measured field data still cannot avoid the ‘afterwards’ characteristics completely. Therefore, bridge management departments have no suitable means to predict the occurrence of VIV and cannot take appropriate response measures. Nowadays, the general management method of VIV is to adopt temporary traffic control measures according to the vibration situation on the spot when VIV has occurred, which lacks rationality and cannot meet the needs of safe bridge operation. Fortunately, most of China's long-span bridges have a structural health monitoring system installed, which generally includes wind field and vibration acceleration monitoring systems. This provides the possibility for us to study the wind field and wind effect characteristics of the bridge based on long-term field-measurement data. In this study, a large amount of wind speed data and vibration acceleration data, collected from a long-span suspension bridge at the East China Sea from 2014 to 2017, are processed and analyzed. Firstly, the main wind field characteristic parameters are studied to determine whether they are related to the occurrence of VIV. Then, through the analysis of the real-time vibration acceleration data of the bridge, the vibration energy concentration coefficient and root mean square (RMS) value of vibration acceleration are found to have high specificity and can be determined as indexes to identify whether VIV occurs. Finally, four index values are utilized to establish a prediction model to warn about the occurrence of VIV, including the wind speed, wind direction, vibration energy

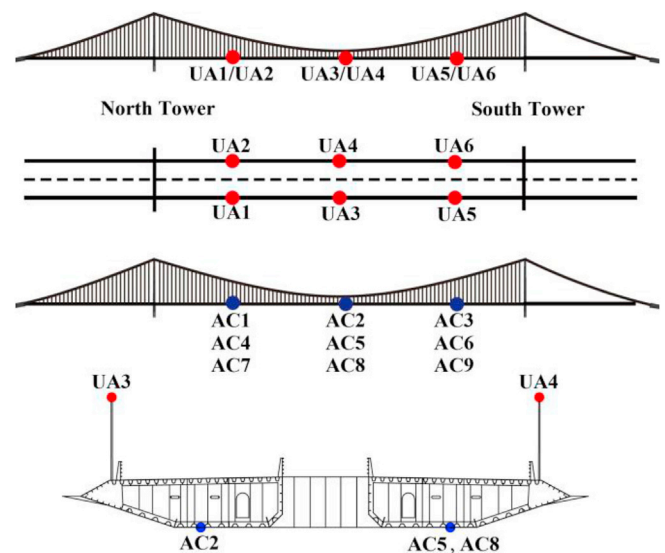


Fig. 1. The layout of the partial sensors installed on the Zhoushan Sea Crossing Bridge.

concentration coefficient, and RMS of vibration acceleration. This warning system is applied in practical engineering to effectively send out a warning signal at the early stage of VIV. Then, the bridge management department can take emergency measures to ensure the safety of the drivers and reduce the economic losses caused by the VIV.

2. Engineering research background

2.1. Introduction of monitoring system

The Zhoushan Sea Crossing Bridge is one of the longest sea-crossing bridges in the project linking the Zhoushan mainland and islands. The bridge extends from north to south, with Brochure island in the north and Jintang island in the south. The hydrogeological conditions of the bridge area are complicated and the wind environment is harsh. The bridge is a two-span suspension bridge with a main span of 1650 m and a side span of 578 m. The main span consists of 126 beams, of which 108 standard and 18 special beam sections. The length of each standard type beam segment is 18m and every 3.6m set a transverse plate separator. The standard type beam is flat streamlined and separate double box section with its total width 36m, height 3.5m and the width of each enclosed steel box is 14m. In addition, the wind fairing is designed to decrease the resonance of the deck structure caused by wind. The transverse distance of the two separate double box section is 6m and are connected by a closed box and a I-shaped connection beam (generally located at the hoisting point). The transverse connecting box girder and I-beam are 3.51m in height and 3.6m in width. The tower is 211 m high, the rise-to-span ratio is 1/10, and the space between each cable is 18 m (Ma et al., 2018). A full set of structural health monitoring systems was installed on the bridge during the construction stage, including systems monitoring the environmental temperature, wind speed, wind direction, strain, vibration acceleration, displacement, and other structural response parameters. The accumulated monitoring data provides an effective foundation for the research presented in this paper.

To explore the relationship between the wind field characteristics (such as the wind speed or wind direction) and the occurrence of VIV, six three-dimensional ultrasonic anemometers (UA1-UA6) were installed on the cross-section at 1/4, 1/2, and 3/4 of the main span. In addition, to compare the vibration parameters (such as the vibration frequency or acceleration amplitude) before and after VIV, nine one-way acceleration sensors were mounted at 1/4, 1/2, and 3/4 of the main span (six sensors, AC1–AC6, monitor the vertical vibration acceleration and three sensors,

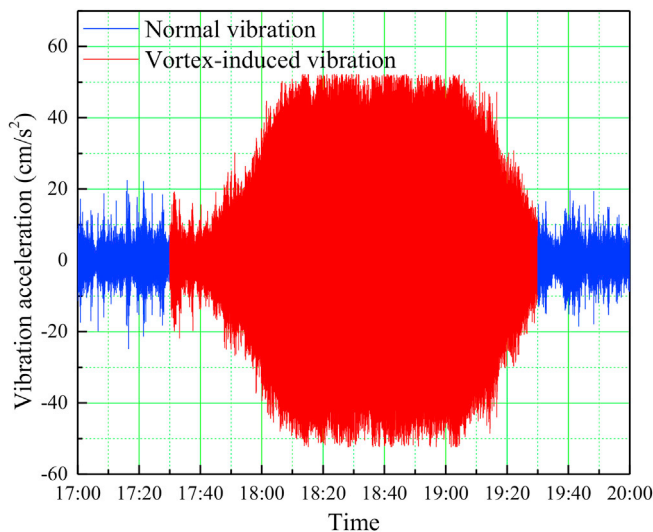


Fig. 2. Acceleration of main beam before and after VIV.

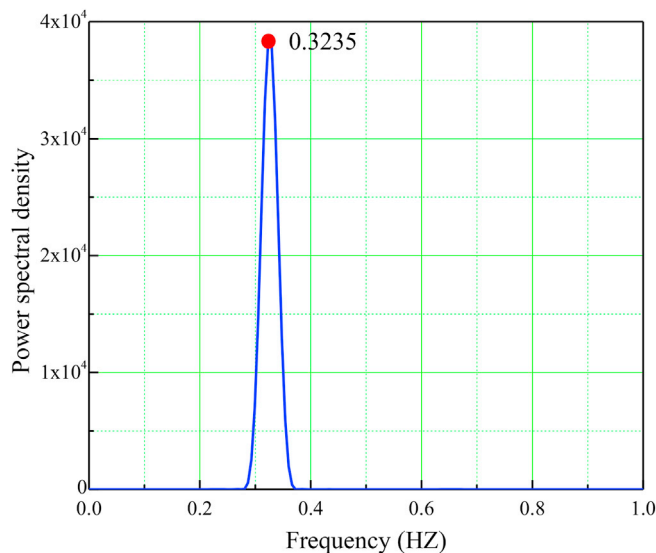


Fig. 3. Vibration power spectral density of the mainbeam during VIV.

AC7–AC9, monitor the lateral vibration acceleration). The layout of the anemometers and acceleration sensors is shown in Fig. 1. The wind field and vibration acceleration data collected from 2014 to 2017 were selected to conduct the detail analysis presented in this paper. By data cleaning, irrelevant and duplicate data are deleted from the original database, missing and abnormal values are filled. Then, the characteristic roots and the cumulative contribution rate of each sensor are calculated by principal component analysis to compress data. According to the analysis results, when UA1, UA3 and UA5 are selected, the retention of original data features has exceeded 90%. So, UA1, UA3 and UA5 are selected for analysis in this paper to obtain the wind field parameters to identify VIV. For acceleration sensors AC1–AC9, it was found that the monitoring data of these sensors located at different locations of the same

longitudinal section had certain correlation. Therefore, those acceleration sensors installed on the same longitudinal section, only one sensor was chosen to be analyzed to improve the analysis efficiency. Finally, the monitoring data of three anemometers (UA1, UA3, and UA5) and three vibration acceleration sensors (AC1, AC3, and AC5) were utilized to establish the VIV prediction model.

2.2. General characteristics of VIV

When air flows around a bridge, vortex shedding will be produced around the main girder. When the vortex-shedding frequency is equal to the natural frequency of the bridge, the main girder will incur significant resonance vibration. Therefore, VIV is essentially an amplitude-limiting flow–solid coupling vibration. To determine the general vibration response characteristics of the bridge during the period of VIV, one of the VIV events, which occurred in August 4, 2014, was taken as an example to carry out detailed analysis. The data analysis period is from 17:00 to 20:00 on August 4, 2014, and the vibration acceleration value of AC5 is shown in Fig. 2. The amplitude of the main girder increases sharply when VIV occurs. It is obvious that the vibration acceleration value of the girder is less than 20 cm/s^2 at the beginning. Then, from 17:30 to 18:00, it quickly increases to over 30 cm/s^2 . During the period 18:10–19:10, the vibration is strong and the vibration acceleration value is kept at about 50 cm/s^2 . The VIV lasted about 2h, from 17:30 to 19:30, and the vibration data collected in that period was used for spectrum analysis. The spectrum analysis result is shown in Fig. 3. The main beam is dominated by only one mode in this period and the vibration frequency is 0.3235 Hz, which is consistent with the theoretical analysis of traditional VIV.

From the above analysis, it is obvious that the vibration acceleration value of the bridge has a high specificity during the period of VIV. Therefore, it is possible for us to assess whether the VIV occurs according to the RMS value of the vibration acceleration. Further, the time and number of VIV events that occurred from 2014 to 2017 have been recorded by the bridge management department. According to the records, a total of 100 VIV events have occurred from 2014 to 2016, including 26 in 2014, 35 in 2015, and 39 events in 2016. To obtain the basic characteristics of VIV, the distribution of the resonance frequency and duration of the 100 VIV events were analyzed. The preliminary data analysis results showed that the resonance frequency of 92 events was in the sixth-order vibration mode (0.327 Hz), of 6 events in the eighth-order mode (0.435 Hz), and only one event in the fourth- (0.230 Hz) and the seventh-order (0.434 Hz) modes. Both the sixth- and eighth-order vibration modes of the bridge are the vertical bending of the main girder and are shown in Fig. 4. Compared with the 92 VIV events under the sixth-order mode, the monitoring data of VIV events under other modes is too small to be of statistical significance. Furthermore, according to the preliminary analysis, only RMS value of the vibration acceleration under different modes has a certain difference, but it does not affect the recognition parameter. Therefore, all the VIV events under all different modes are analyzed together without distinction in this paper. In addition, the main goal of the paper is to recognize VIV as early as possible. However, how long is it reasonable to recognize VIV after its occurrence? Then, the duration of the 100 VIV events were statistically analyzed. The results show that 10% of VIV events lasted 0–20 min, 26% lasted 20–60 min, 51% lasted 60–120 min, and 13% lasted more than 120 min. Therefore, if a VIV event can be recognized within 20 min after its occurrence, this will guarantee 90% of VIV events can be recognized and give early warning to the management department of the bridge. So, the



Fig. 4. The sixth and eighth vibration modes of the Zhoushan Sea Crossing Bridge.

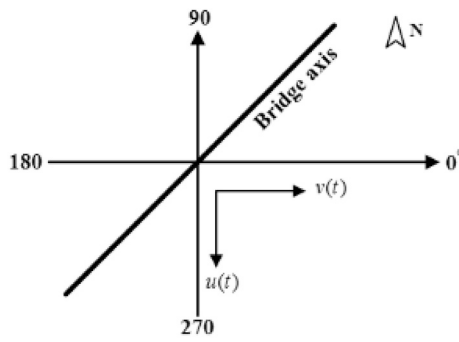


Fig. 5. Schematic of wind speed and direction of the three-directional ultrasonic anemometers.

main goal of the prediction model is to recognize a VIV within 20 min after its occurrence. Then, the management department of the bridge may have enough time to take appropriate management measures to ensure the safety of running vehicles.

3. Study of wind field characteristic parameters of VIV

Apparently, VIV is caused by wind loads. Hence, the occurrence of VIV must be related to the characteristics of the wind field, such as the wind speed, wind direction, turbulence, or other factors. Therefore, it is necessary to study the characteristics of the wind field during VIV. In this section, common wind field characteristic parameters used in wind engineering—including the average wind speed, wind direction, the

degree of turbulence, and gust factor—are calculated to determine whether these parameters have specificity when VIV occurs. Through in-depth analysis of the above parameters, the correlation between these four wind field parameters and the occurrence of VIV can be explored, which provides a data base for the VIV prediction model.

3.1. Wind speed and wind direction

The wind speed measured by the three-directional ultrasonic anemometers installed on the bridge is defined as $u(t)$, $v(t)$ and $w(t)$. The definition of wind speed $u(t)$, $v(t)$ is shown in Fig. 5. The angle of the bridge axis is about 45° . Taking 10 min as the basic calculation time interval, it is assumed that the horizontal average wind speed and its corresponding wind direction are U and Φ , respectively. Then, the measured data are analyzed according to the following formula:

$$U = \sqrt{u(t)^2 + v(t)^2} \quad (1)$$

$$\cos\Phi = \frac{u(t)}{U} \quad (2)$$

By collecting the data of the 100 VIV events from 2014 to 2016 and setting each 10 min interval as the calculation condition, a total of 1542 calculation conditions were studied for the VIV of the bridge. Similarly, by setting 10 min as the calculation condition, 4626 (1542×3) normal calculation conditions without VIVs were randomly selected from the bridge monitoring data. The average wind speed and its statistical information during VIV and the normal vibration of the bridge were calculated and the calculation results are shown in Fig. 6.

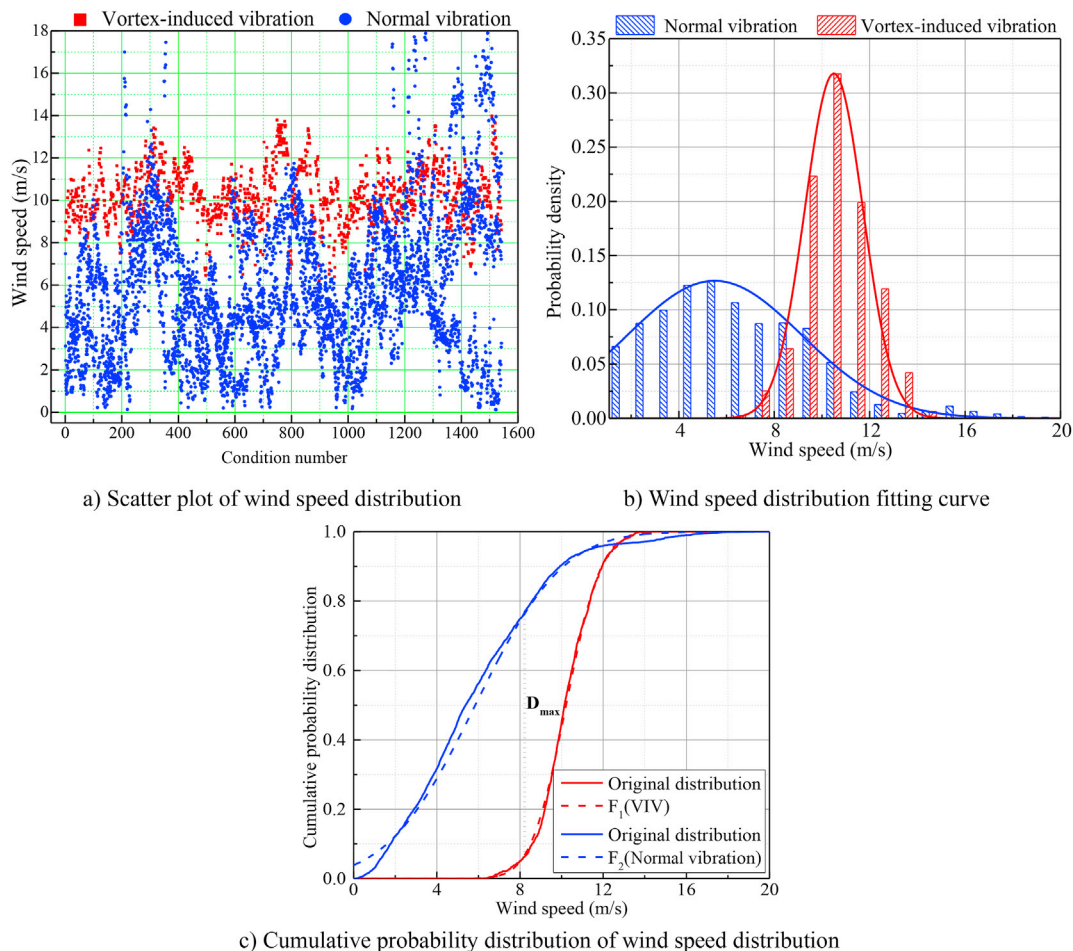


Fig. 6. Comparison of wind speed distribution for VIV and normal vibration.

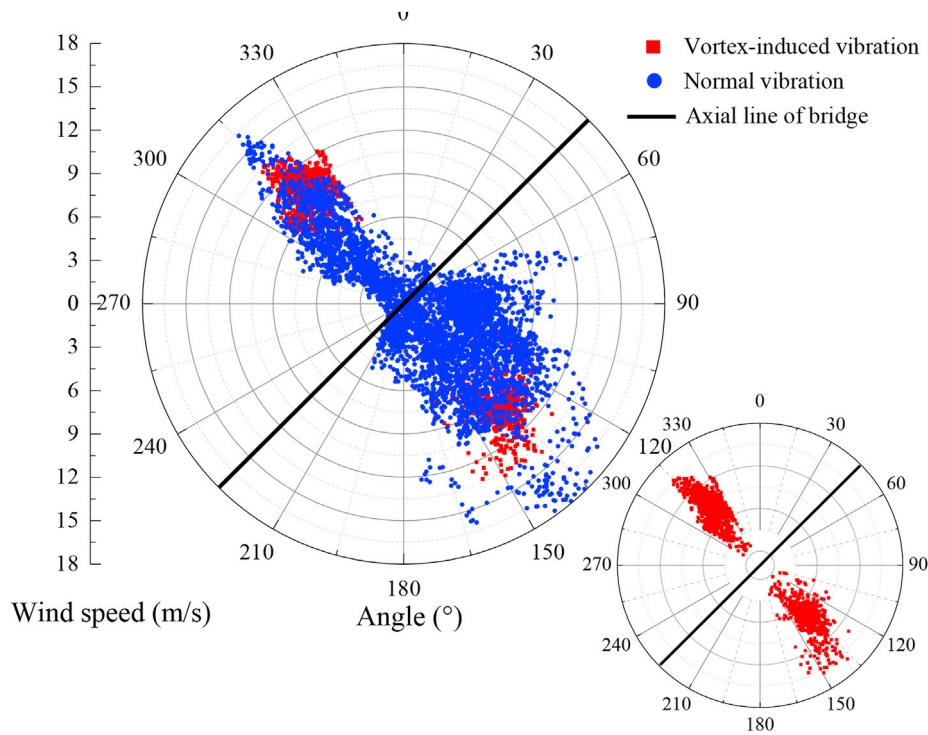


Fig. 7. Comparison of wind direction for VIV and normal vibration.

As shown in Fig. 6 a), the red scatter is the distribution of the average wind speed measured by UA3 when VIV occurred. Through statistical calculations, the horizontal wind speed distribution was found to range from 6.2 m/s to 13.9 m/s, the mean value was 10.2 m/s, and the standard deviation was 1.3. The blue scatter in Fig. 6 a) is the distribution of the average wind speed measured by UA3 under normal vibration. The calculated wind speed distribution ranges from 0.1 m/s to 19.3 m/s, the average value is 5.8 m/s, and the standard deviation is 3.3. It can be seen from Fig. 6 b) that the average wind speed distribution in the case of VIV is significantly different from that in the case of normal vibration. In addition, the wind speed distribution curve has no intersection area in the range 0–6 m/s. By comparing the wind speed in these two cases, it is obvious that VIV may occur only when the average wind speed is within a certain range. It can be inferred that the average wind speed may be an important characteristic parameter that can be used to determine whether VIV occurs. The K-S test was applied in the analysis according to formulas (3)–(5) to quantitatively analyze whether the average wind speed is specific when VIV occurs.

$$Z = \max |D_i| \sqrt{\frac{n_1 n_2}{n_1 + n_2}} \quad (3)$$

$$D_i = F_1(X_i) - F_2(X_i) \quad (4)$$

$$Z_0 = f(\alpha) \sqrt{\frac{n_1 + n_2}{n_1 n_2}} \quad (5)$$

Where Z is the test statistics, D_i is the difference of the two distribution, X_i is wind speed, F_1 and F_2 are the cumulative probability distribution function of the two samples, n_1 and n_2 are the sample size, Z_0 is the threshold to determine whether the difference is significant, α is significant level and $f(\alpha)$ can be referenced as 1.36 when α is defined as 5%. The test statistics values of the two groups of data are compared with the threshold Z_0 to determine whether the difference is significant. If $Z > Z_0$, there is significant difference between the two groups of data. Otherwise, there is no significant difference between the two groups of data and

these two distributions are consistent. As is shown in Fig. 6 c), the cumulative probability distribution of wind speed distribution and its fitting curves F_1, F_2 are calculated. Then, by calculating the parameter D_i of the two groups of data, $\max |D_i| = 0.71$ when the wind speed is 8.36 m/s and $Z = 24.14 > Z_0 = 0.04$. It is obvious that there is a significant difference between the two groups of data. Therefore, it can be determined that the average wind speed during VIV is significantly different from that under normal vibration. The average wind speed can be used as an index to determine whether VIV may occur. The same calculation method can be used to obtain the average wind speed distribution of UA1 and UA5 when VIV occurs.

The wind direction refers to the angle between the prevailing wind direction and the horizontal plane of the wind and indicates the effect of natural wind blowing on the bridge, which is one of the important parameters of the wind field. Equation (2) can be used to calculate the wind direction. Fig. 7 shows the relationship between the wind speed and the wind direction, which is obtained on the basis of the data collected by the UA3 sensor from 2014 to 2016.

It can be concluded from Fig. 7 that the distribution range of the wind direction during the period of VIV is 120–150° and 300–330° (with a difference of 180°) and perpendicular to the bridge axis. Owing to the influence of the topography of the bridge site, the wind direction under normal vibration is mostly close to that of VIV. However, other wind directions still exist that range from 0 to 360°. That is, the wind direction has specificity during the period of VIV. Therefore, only when the wind direction is in a certain range (120–150° and 300–330° for UA3), VIV may occur on the bridge. The wind direction can also be taken as an index to determine whether VIV occurs.

3.2. Turbulence parameter analysis of wind field

The turbulence intensity and gust factors are two important parameters to describe the turbulent characteristics of a wind field. Therefore, this section describes the analysis of the correlation between turbulence parameters and the occurrence of VIV. The turbulence intensity, and the gust factor can be calculated by formulas (6), (7), and (8), respectively.

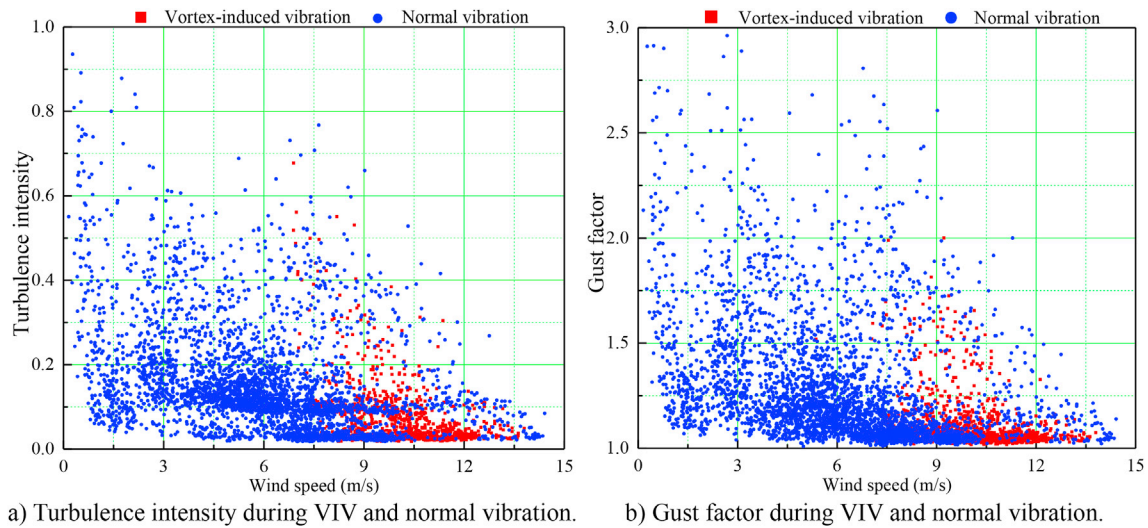


Fig. 8. Comparison of turbulence parameter for VIV and normal vibration.

$$u_f(t) = u(t)\cos\Phi + v(t)\sin\Phi - U \quad (6)$$

$$I_U = \frac{\sigma_U}{U} \quad (7)$$

$$G_U(t_g) = \frac{\max(\overline{u(t_g)})}{U} \quad (8)$$

$\mu_f(t)$ is the fluctuating wind speed, σ_u is the standard deviation of the fluctuating wind speed, U is average wind speed, Φ is wind direction, t_g is gusty duration and is set as 3s in this study. Fig. 8 shows the comparison of the turbulence intensity and gust factor during VIV with those during normal vibration. It can be drawn from the preliminary comparison of the data points in the figure that the turbulence intensity and gust factor of the bridge during the period of VIV are basically overlapped with those obtained during normal vibration. It is difficult to find obvious characteristics from the distribution diagram of these two parameters directly. The statistical characteristics of the turbulence intensity and gust factor of the bridge during the period of VIV and under normal vibration were calculated. The turbulence intensity was distributed in the range 0–0.7, with a mean of 0.08 and a standard deviation of 0.07 during the period of VIV. Under normal vibration, the distribution range of the turbulence intensity parameters was 0–1, the mean value was 0.13, and the standard deviation was 0.14. The gust factor parameters were distributed in the range 1–2, with a mean of 1.12, and a standard deviation of 0.16 during the period of VIV. Under normal vibration, the distribution range of the gust factor was 1–3, the mean value was 1.3, and the standard deviation was 0.36. By calculating the K-S test statistical parameters of these two parameters according to formulas (3)–(5), the Z values of the turbulence intensity and gust factor were equal to 0.017 and 0.029, respectively. Because Z_0 is equal to 0.04, the Z values of both the turbulence intensity and the gust factor are smaller than Z_0 . Therefore, it can be determined that the distributions of the turbulence intensity and gust factor of VIV are similar to those of the normal vibration. The turbulence intensity and gust factor cannot be utilized as indexes to establish the VIV prediction model.

The in-depth extraction of wind field data is described in this section. The average wind speed, wind direction, turbulence intensity, and gust factor were calculated to analyze their correlation with VIV. The average wind speed was determined as an index to predict the occurrence of VIV according to differences significant contrast analysis. However, the turbulence density and gust factor have no obvious characteristics when VIV occurs. In addition, VIV may occur only when the wind direction is mainly distributed in 120–150° and 300–330°, and the wind direction is

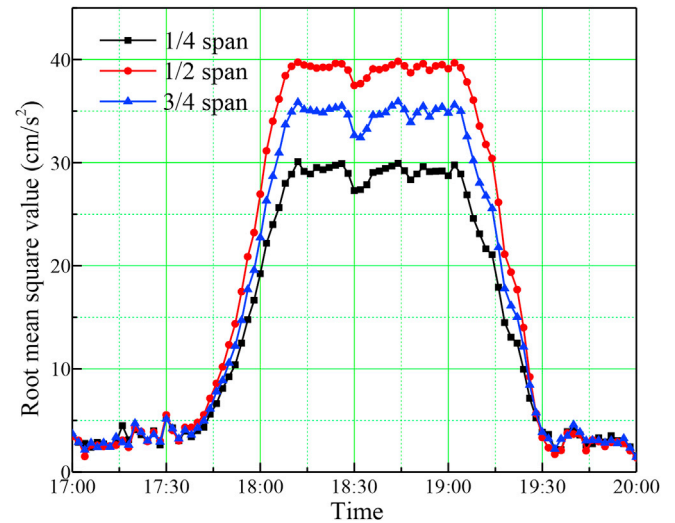


Fig. 9. RMS value of the vibration acceleration of the main bridge before and after VIV.

basically perpendicular to the bridge axis. Therefore, the wind direction is also taken as an index to predict the occurrence of VIV.

4. Analysis of vibration characteristics of VIV

According to wind engineering theory, VIV is essentially a special vibration state of the bridge. Therefore, the vibration response data of the bridge must be analyzed to describe VIV. Firstly, the vibration acceleration value of the bridge has a high specificity during the period of VIV. Moreover, according to research results of wind-induced vibration theory, VIV is essentially a resonance phenomenon between the bridge structure and the vortex-shedding frequency. VIV occurs when the bridge vibrates at a single frequency. Therefore, the vibration frequency of the bridge should also have high specificity during the period of VIV. In view of this, the vibration acceleration data were studied to determine the index of the VIV prediction model.

4.1. Characteristics of the RMS value of the vibration acceleration

In the present study, the parameter data describing the intensity of the vibration are the absolute and the RMS value of the vibration ac-

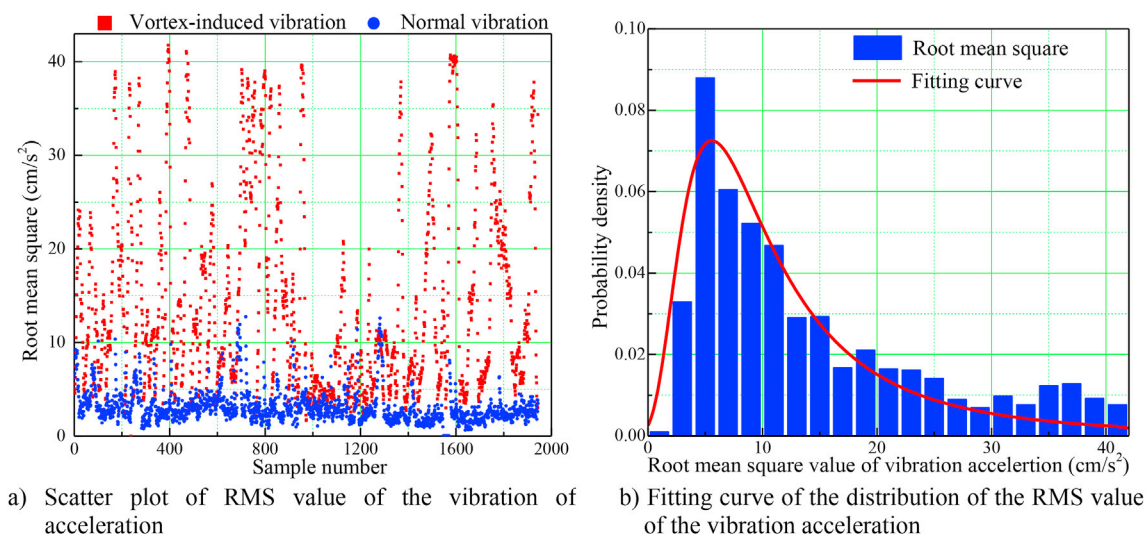


Fig. 10. Comparison between RMS values of the vibration for VIV and normal vibration.

celeration. Because the RMS value can more fully express the varying intensity of the vibration acceleration, the RMS value of the vibration acceleration data was calculated in this study. To fully grasp the characteristics of VIV, the analysis interval was set as 2 min. Then, the variation of the RMS value of the vibration data of the bridge from 17:00 to 20:00 on August 4 in 2014 was analyzed. It can be drawn from Fig. 9 that the RMS value is below 5 cm/s^2 under normal vibration. During VIV, the RMS value first presents an increasing trend, then is maintained stable, and decreases at the end. During the period of VIV, the maximum RMS value reaches 40 cm/s^2 . Moreover, an obvious plateau is presented in the time diagram, which is the same as that observed in the variation of the vibration acceleration value. The vibration characteristics at 1/2 of the span of the main bridge during VIV are similar to those obtained at 1/4 and 3/4 of the span of bridge. Moreover, in terms of start time, end time, and vibration amplitude, the performance is consistent in each section. When VIV occurs on the bridge, the RMS value of the vibration shows the behavior $1/2 \text{ span} > 3/4 \text{ span} > 1/4 \text{ span}$.

Fig. 10 shows a comparison between the vibration RMS values of the main bridge under normal vibration and for the 100 VIV events from 2014 to 2016, obtained by setting the calculation interval as 2 min. According to the data distribution in Fig. 10 a), the RMS value of the main bridge is almost always above 5 cm/s^2 , and some values even reach 40 cm/s^2 during VIV. However, for normal vibration, the variation range of the RMS value is between 0 and 10 cm/s^2 , with most RMS values below 5 cm/s^2 . Therefore, the RMS value has obvious specificity during the period of VIV and can be used as an index to establish the VIV prediction model. That is, when VIV occurs, the RMS value increases rapidly and is greater than a certain threshold. To determine the threshold value, the statistical results of the RMS value of the main beam during the 100 VIV events were calculated. Fig. 10 b) shows the distribution of the RMS value of the 100 VIV events. Through the curve fitting, the RMS value of the vibration acceleration is found to agree with the Generalized Extreme-Value (typeII) Distribution with shape parameter (shape) $k = 0.4127$, $\sigma = 5.4712$, and location parameter (location) $\mu = 8.3346$. According to the fitting distribution function, the distribution range of the RMS value of the vibration acceleration varies between 4 cm/s^2 and 40 cm/s^2 when VIV occurs. The RMS value of the vibration acceleration of 95% of the VIV events is greater than 5 cm/s^2 . It follows that most of the VIV events of the bridge caused strong vibration of the main beam. Therefore, the threshold value is determined as 5 cm/s^2 according to the distribution of the RMS value. When the RMS value presents an increasing trend over a period of time and is greater than 5 cm/s^2 , there will be a possibility of VIV occurrence on the bridge.

4.2. Vibration energy concentration coefficient

To further explore the characteristic vibration parameters of the main girder during VIV, the spectral analysis of the vibration acceleration data acquired during the period from 17:00 to 20:00 on August 4, 2014 is still used as the example to carry out detailed analysis. By setting the analysis interval as 30 min, six vibration frequencies distribution diagram of the bridge can be obtained through spectrum analysis. The recognition results are shown in Fig. 11, which clearly shows the variation of the vibration frequencies of the main girder of the bridge from normal vibration to VIV and back to normal vibration. Fig. 11 a) shows the power spectrum of the bridge under normal vibration from 17:00 to 17:30. The bridge is vibrating under the coupling of a variety of vibration modes simultaneously and the corresponding vibration frequency is 0.1038 Hz, 0.2319 Hz, and 0.3296 Hz. In addition, the maximum energy in the spectrum is almost the same as the magnitude of the second mode. During the period of 17:30 to 18:00, the dominant vibration frequency is 0.3296 Hz and the vibration energy of the other modes is very small. However, in the vicinity of 0.1 Hz and 0.2 Hz, two modes with smaller vibration energy still exist. At this stage, a transformation of multimodal vibration to single-mode vibration occurs. During the period 18:00–19:00, besides the main mode at 0.3296 Hz, it can be drawn from Fig. 11 c) and d) that the other frequency values in the spectrum of the power spectral density are almost 0. That is, the bridge vibrates at a single frequency. The vibration frequency in the period from 19:00 to 20:00 is shown in Fig. 11 e) and f), which resembles the inverse process of 17:00 to 18:00, and the vibration of the bridge is transformed from a single-mode to a multimode coupled vibration.

According to the above analysis, the normal vibration of the bridge is a multimode coupled vibration. Therefore, there are a number of peaks in the power spectral density diagram. However, the main beam only vibrates at the resonance frequency when VIV occurs. Then, only one peak is dominant in the power spectrum diagram, and the energy of the other frequencies is close to 0. This characteristic of VIV can be used as a parameter to determine whether VIV occurs. However, to effectively predict the occurrence of VIV, the most important thing is to extract an index that presents the vibration characteristics of VIV. Based on the results of the power spectrum analysis, the most pronounced difference between VIV and normal vibration is the value of the frequency energy. The maximum power spectral density in the spectrum analysis of the main girder is defined as P_1 and the second-highest power spectral density as P_2 . During VIV, only one frequency corresponds to the main vibration mode ($P_2 \approx 0$). Therefore, the vibration energy concentration coefficient (P_2/P_1) is used as the parameter that characterizes the power

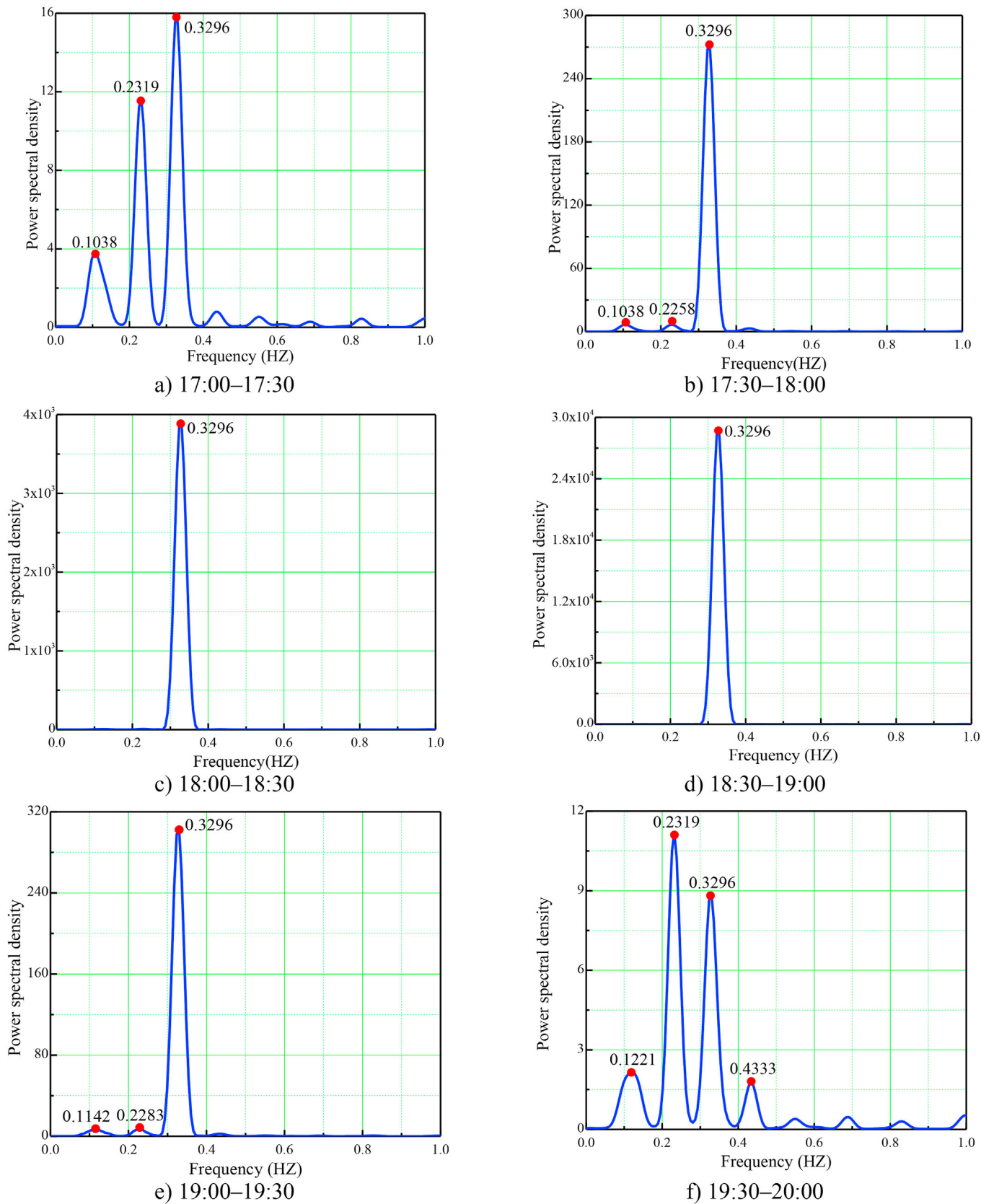


Fig. 11. Vibration spectrum of the main girder.

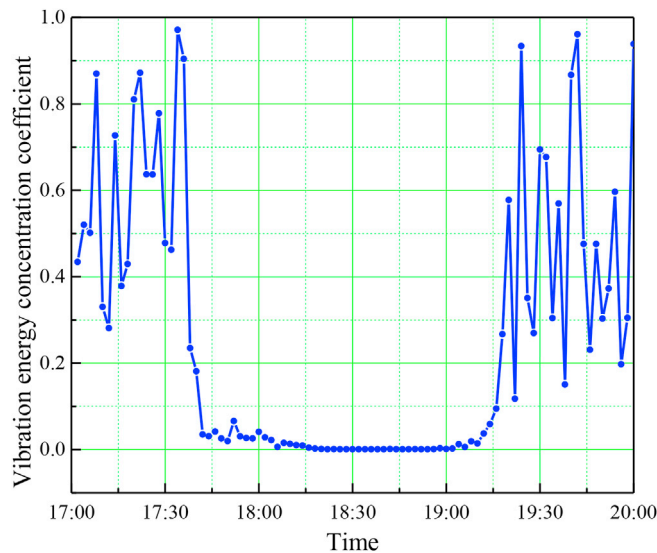


Fig. 12. The variation of vibration energy concentration coefficient from 17:00 to 20:00 on August 4 in 2014.

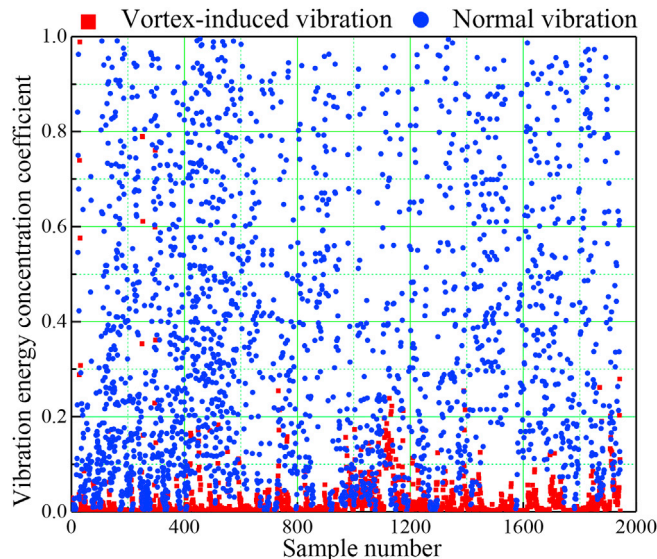


Fig. 13. Vibration energy concentration coefficient of VIV and normal vibration.

spectrum distribution between VIV and normal vibration, which is defined by the following formula:

$$\begin{aligned} P_2/P_1 &\approx 0 && \text{VIV} \\ 0 &\ll P_2/P_1 < 1 && \text{Normal vibration} \end{aligned} \quad (9)$$

To obtain the variation of the vibration energy concentration coefficient, the parameter is calculated during the period from 17:00 to 20:00 on August 4, 2014 and the calculation interval is set as 2 min. As is illustrated in Fig. 12, the coefficient is less than 0.1 (close to 0) during the period 17:45–19:15 and greater than 0.2 during all other times, which confirms the above analysis. The variation of the vibration energy concentration coefficient of the main girder can be used as an important index for the prediction of VIV. To obtain an accurate threshold of the index, the P_2/P_1 value of 100 VIV events and 300 normal vibrations was calculated. It can be drawn from Fig. 13 that most of the values of the vibration energy concentration coefficient in normal vibration are between 0.1 and 1.0, with only a few values ranging from 0 to 0.1. However, during VIV, the value of the vibration energy concentration

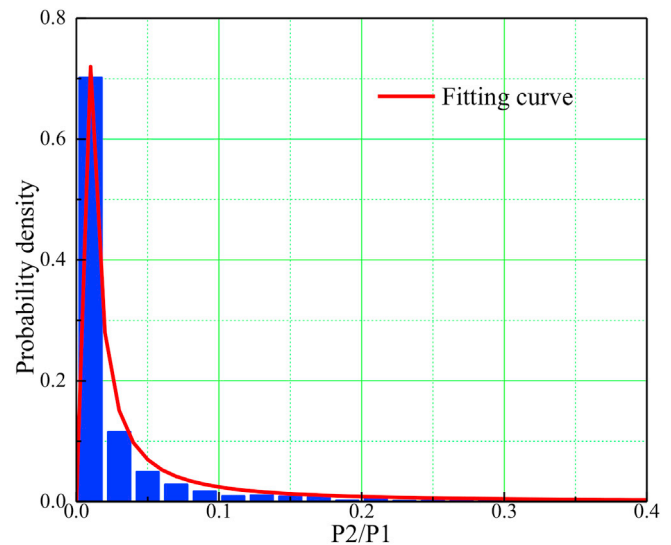


Fig. 14. Statistical diagram of P_2/P_1 during VIV.

coefficient is basically between 0 and 0.1, except for a small number of sample points. The data points that do not conform to the above laws may originate from the mixing of VIV and normal vibration due to errors in the recording of the occurrence time of VIV.

Fig. 14 shows the vibration energy concentration coefficient calculated from the 100 VIV events recorded from 2014 to 2016 and the fitting function. It can be determined that the vibration energy concentration coefficient corresponds to the Boolean function distribution (Burr), with proportional parameter (scale) $\alpha = 0.0013$, shape parameter 1: $c = 2.9475$, and shape parameter 2: $k = 0.1762$. As shown in Fig. 15, P_2/P_1 varies from 0 to 0.4 and more than 98% of P_2/P_1 values are smaller than 0.1. Therefore, the threshold of the vibration energy concentration coefficient that can be used to determine whether VIV occurs is set as 0.1. That is, it can be considered that the bridge has a high probability of VIV when P_2/P_1 is less than 0.1.

According to the vibration amplitude and spectral analysis of 100 VIV events, the vibration characteristics of VIV on the long-span suspension bridge are further explored. While the bridge vibrates under normal conditions, the RMS value ranges from 0 to 10 cm/s^2 and most RMS values are below 5 cm/s^2 . However, during VIV, the RMS value is basically greater than 5 cm/s^2 and even reaches 40 cm/s^2 . Therefore, when the RMS value reaches above 5 cm/s^2 and presents an increasing trend over a period, it can be used as an index to predict the occurrence of VIV. In addition, the normal vibration of the bridge is a multimodal coupled vibration, while VIV is a single-frequency vibration under the action of low wind speed. The vibration energy concentration coefficient of the power spectral density corresponding to the maximum power spectral density (P_1) in the spectrum analysis and the second-highest power spectral density (P_2) can characterize the changes in the vibration modes of the bridge. The value of vibration energy concentration coefficient (P_2/P_1) can also be used as an index to assess VIV. That is, the bridge is considered to have incurred VIV when the value of the vibration energy concentration coefficient (P_2/P_1) is substantially smaller than 0.1.

5. Establishment of VIV prediction model

Through the above analysis, four parameters, including the average wind speed, the wind direction, the RMS value, and frequency of the vibration energy concentration coefficient, are chosen as the indexes to determine the occurrence of VIV. Of all these parameters, the wind speed and wind direction are important parameters to describe the characteristics of the wind field at the bridge site. Both the RMS value of the vibration acceleration and the vibration energy concentration coefficient

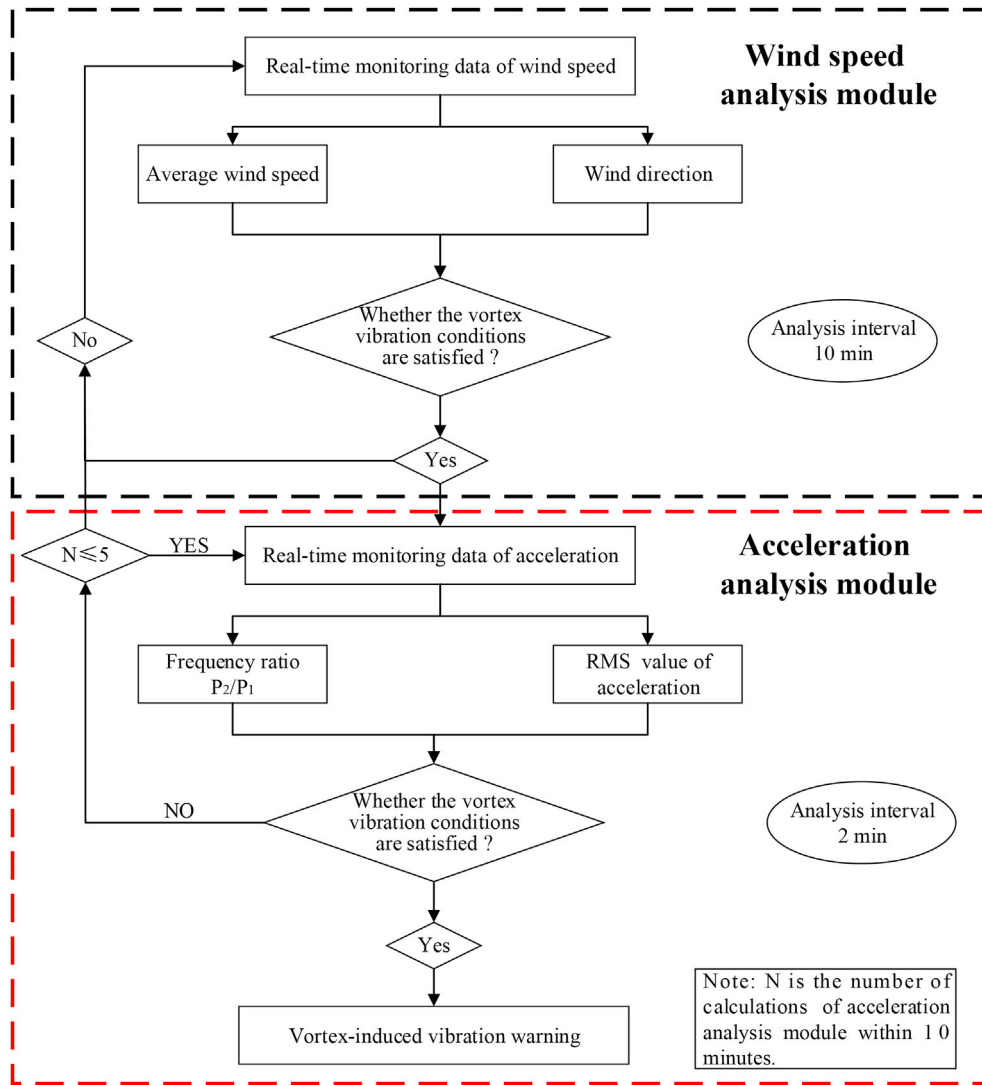


Fig. 15. Flow chart of dynamic VIV prediction model.

(P_2/P_1) of the power spectrum are important vibration characteristics that can be used to distinguish VIV from the normal vibration of the bridge. Based on these four characteristic parameters, the dynamic prediction model of VIV for the Zhoushan Sea Crossing Bridge is established in this section. The prediction model was designed to quickly and effectively identify and predict VIV using these four indexes, which were derived from real-time monitoring data. The prediction model needs to send warning signals as soon as possible before the bridge vibrates strongly under the action of VIV. Then, the bridge management department must adopt measures, such as traffic restriction or speed limits, to reduce the impact of VIV on driver safety.

The framework of the prediction model is shown in Fig. 15. The prediction model is mainly divided into two parts. One is the wind field data analysis module and the other is the vibration acceleration data analysis module. The wind field data module is used to assess whether the average wind speed and wind direction meet the requirements of VIV occurrence. If the two indexes of the wind field are satisfied, then vibration acceleration analysis is carried out. Otherwise, the real-time monitoring data of the wind field in the next time interval are analyzed. Based on the analysis of the vibration acceleration data, the RMS value and the vibration energy concentration coefficient are obtained to determine whether VIV has happened. If the wind speed analysis module meets the threshold value of VIV, and the acceleration analysis module does not meet the threshold value, the acceleration

Table 1
Countermeasures to different VIV events.

Warning level	Amplitude of vibration acceleration(m/s^2)	Driver perception	Countermeasures
I	$A_m \leq 0.08$	Slight	No warning
II	$0.08 < A_m \leq 0.39$	Acceptable	Constant attention
III	$0.39 < A_m \leq 1.17$	Uncomfortable	Warning vehicles to slow down
IV	$A_m > 1.17$	Intolerable	Prohibit the passage of vehicles

analysis module will be continuously carried out at an interval of 2 min, until the recognition of VIV or the number of calculations is equal to 5 (10 min). After the VIV is identified, different countermeasures will be taken according to the maximum acceleration value and driver perception during the VIV. Specific measures are shown in Table 1 which is developed based on research results of Chen and Huang (2015).

5.1. Index scoring principle of VIV prediction model

Based on the analysis of the data measured by UA3, the wind speed mainly ranges from 6 m/s to 14 m/s when VIV occurs. Moreover, the wind direction is mainly distributed in the ranges of 120°–150° and

Table 2

The threshold value of the average wind speed and wind direction during VIV.

Sensor number	UA1	UA3	UA5
Wind speed range	7.17–13.30	6.21–13.96	5.57–12.07
Wind direction range	120–150/300–330	120–150/300–330	308–337

300°–330°, which is perpendicular to the axis of the bridge. Usually, the calculation interval is set as 10 min, according to the bridge's wind engineering specification. Therefore, for the wind field data used to establish the prediction model in this study, 10 min was also set as the analysis interval. According to equations (1) and (2), the average wind speed and wind direction of UA1 and UA5 can be calculated. Table 2 shows the threshold values of the average wind speed and the wind direction for each sensor when VIV occurs.

To fully express the wind field indexes (average wind speed and wind direction) of each sensor in the VIV prediction model, index scoring rules are introduced, which are shown in formula (10).

$$\begin{aligned}
 \alpha_i &= 1 && \text{The average wind speed meets the threshold value of VIV.} \\
 \alpha_i &= 0 && \text{The average wind speed does not meet the threshold value of VIV.} \\
 \beta_i &= 1 && \text{The wind direction meets the threshold value of VIV.} \\
 \beta_i &= 0 && \text{The wind direction does not meet the threshold value of VIV.}
 \end{aligned} \quad (10)$$

The parameters $\alpha_i, \beta_i (i = 1, 3, 5)$ represent whether the average wind speed and wind direction of sensor i meet the threshold value of VIV. When the wind field data meet the threshold values of wind speed and wind direction, it can be concluded that the wind field data are in accordance with the characteristics of VIV. That is, the occurrence possibility of VIV is very high under such wind field conditions at this point. Then, further analysis is carried out by the acceleration data analysis module. Otherwise, the wind field data of the following 10 min are continuously calculated.

When input in the vibration acceleration data analysis module, the RMS value and the vibration energy concentration coefficient are determined as indexes used to evaluate the occurrence of VIV. On the basis of the above analysis, the threshold value of VIV for each sensor (AC1, AC3, AC5) is 5 cm/s^2 . Moreover, the threshold value of the vibration energy concentration coefficient for each sensor is determined to be 0.1. Because the duration of VIV is generally from 30 min to 120 min, the prediction model should recognize the occurrence of VIV as quickly as possible. Furthermore, the sampling frequency of the vibration acceleration sensors of the bridge is 50 Hz, and a data volume of 2 min meets the requirements of the spectrum analysis. Therefore, the analysis time interval in the acceleration analysis module is set as 2 min to improve the data analysis speed. Similarly, two parameters are introduced to represent whether the frequency ratio and the RMS value meet the threshold value of VIV. The index scoring rules are expressed by the following formula.

$$\begin{aligned}
 \mu_i &= 1 && \text{The RMS value meets the threshold value of VIV.} \\
 \mu_i &= 0 && \text{The RMS value does not meet the threshold value of VIV.} \\
 v_i &= 1 && \text{The vibration energy concentration coefficient meets the threshold value of VIV.} \\
 v_i &= 0 && \text{The vibration energy concentration coefficient does not meet the threshold value of VIV.}
 \end{aligned} \quad (11)$$

The parameters μ_i and $v_i (i = 1, 3, 5)$ represent the scores of the RMS value and the vibration energy concentration coefficient of each sensor, respectively. When both the wind field data and the vibration acceleration data meet the threshold value of the four indexes, it is very likely that VIV will occur on the bridge. In the dynamic VIV prediction model, it is assumed that all sensors have the same weight coefficient in the process

of determining whether VIV occurs. According to the above analysis, it is obvious that three wind speed sensors need to meet the thresholds of two indexes (average wind speed and wind direction) to assess the occurrence of VIV. Therefore, the number of parameters of the wind field data module is $2 \times 3 = 6$. The formula used to determine whether VIV will occur in the wind field data analysis module is as follows.

$$\sum_i \alpha_i + \sum_i \beta_i = 6 \quad (i = 1, 3, 5) \quad (12)$$

When the average wind speed and wind direction meet the threshold value of VIV occurrence (formula (11)), it is necessary to further analyze the other two indexes (RMS value and vibration energy concentration coefficient). Similarly, three vibration acceleration sensors must satisfy the threshold of the RMS value and the vibration energy concentration coefficient to assess the occurrence of VIV. Therefore, the number of parameters of the vibration acceleration data module is also $2 \times 3 = 6$. The formula used to determine whether VIV will occur in the acceleration data analysis module is as follows.

$$\sum_i \mu_i + \sum_i v_i = 6 \quad (i = 1, 3, 5) \quad (13)$$

If the analysis results of the vibration acceleration data do not satisfy formula (13), the prediction analysis returns to acceleration analysis module of the next 2 min. If formula (13) is not satisfied until 5 times of calculation (10 min), the VIV prediction model will re-enter wind speed analysis module. Based on the monitoring data of the bridge in 2017, the VIV prediction model was used to identify VIV events during that period. Then, the recognition results of the VIV prediction model were compared with the VIV events recorded by the management department in 2017. 36 VIV events were recorded in 2017 by the management department and 21 events were identified by the VIV prediction model. In addition, three normal vibrations were identified as VIV events. Therefore, the recognition rate is 58% and the misrecognition rate is 8%. It is obvious that both the recognition rate and the misrecognition rate are very low. Hence, the wind field and acceleration data of VIV events that were not recognized were further analyzed in detail. The four indexes, including the average wind speed, wind direction, RMS value, and vibration energy concentration coefficient, were calculated. The results showed that it is unlikely to simultaneously satisfy the criteria of six parameters of wind field data and six parameters of vibration acceleration data during VIV. In other words, even if VIV occurs, not all sensors meet the thresholds of all parameters. The discrimination mechanism of the VIV prediction model needs further optimization.

5.2. Optimization of discriminant mechanism of VIV prediction model

It can be concluded from the above analysis that only some of the sensors may meet the threshold value of some parameters when VIV occurs. Therefore, if the prediction model requires that all sensors meet

all parameter thresholds, the discriminant mechanism is too strict to recognize all the VIV events. That would explain why both the recognition rate and the misrecognition rate are very low. The number of thresholds needed to assess the occurrence of VIV becomes the key parameter of the prediction model. Therefore, optimization analysis of the discriminant mechanism was carried out and the theoretical formula of the optimization algorithm is as follows.

Table 3

Detailed VIV identification results based on the monitoring data of 2017.

Recognition time/min	0–8	8–10	10–12	12–14	14–16	16–18	18–20	>20
Recognize number	0	2	3	7	11	4	2	3
Misrecognize number	0	0	0	1	0	1	0	2
Recognition accuracy	/	100%	100%	87.5%	100%	80%	100%	60%
Integrated recognition accuracy	32/36 = 89%							

Table 4

Example analysis of VIV dynamic prediction.

Sensor number	UA1	UA 3	UA 5	Aggregate score	Notes
Average wind speed	6.5	7.9	6.2	$\sum = 5$	Larger than 3 meets the requirements of the occurrence of VIV.
Wind direction	328.3	321.8	325.4		
Scores	$\alpha_1 = 0$ $\beta_1 = 1$	$\alpha_3 = 1$ $\beta_3 = 1$	$\alpha_5 = 1$ $\beta_5 = 1$		
Sensor number	AC1	AC3	AC5	$\sum = 4$	Equal to 4 meets the requirements of the occurrence of VIV.
RMS	4.9	5.8	4.1		
Vibration energy concentration coefficient	0.054	0.006	0.047		
Scores	$\mu_1 = 0$ $\nu_1 = 1$	$\mu_3 = 1$ $\nu_3 = 1$	$\mu_5 = 0$ $\nu_5 = 1$		

$$\begin{cases} \sum_i \alpha_i + \sum_i \beta_i \geq A \quad (i = 1, 3, 5) \\ \sum_i \mu_i + \sum_i \nu_i \geq B \quad (i = 1, 3, 5) \\ g(A, B) \rightarrow 100\% \quad (1 \leq A \leq 6, 1 \leq B \leq 6) \\ f(A, B) \rightarrow 0\% \quad (1 \leq A \leq 6, 1 \leq B \leq 6) \end{cases} \quad (14)$$

where $g(A, B)$ and $f(A, B)$ are the recognition and misrecognition rate, respectively. According to formula (14), it is obvious that the value of both A and B are positive integer and range from 1 to 6. So, there are a total of 36 combinations of values of A and B. Then, based on the wind field and vibration acceleration data measured from 2014 to 2016, all the 100 VIV events occurred from 2014 to 2016 were trial-recognized under different combinations of A and B. The recognition result shows that the recognition model has the highest recognition rate when $A = 3$ and $B = 4$. So, the most reasonable values of A and B can be determined. By optimizing the parameters of formulas (12) and (13), the monitoring data of 2017 was taken as input to test the prediction model. The number of recognized and misrecognized VIV and the recognition accuracy in different time periods are shown in Table 3. According to the VIV identification model shown in Fig. 15, the identification time of VIV is mainly determined by acceleration analysis module. So, the recognition time is also at an interval of 2 min.

Through the optimization of the discriminant mechanism of the VIV prediction model, the recognition rate of VIV increased from 58% to 89% and the misrecognition rate increased from 8% to 11%. Despite a 3% increase in the misrecognition rate, the recognition rate was increased by 31% and was greatly improved. To further illustrate the discriminant mechanism of the VIV prediction model, the analysis of the data collected from 16:30 to 16:40 on August 3, 2017 is provided as an example. The specific analysis results are shown in Table 4. The average wind speed and wind direction parameters of each anemometer during an interval of 10 min are shown in the second and third row of Table 4, respectively. The scores of each sensor are shown in the fourth row. The analysis results of acceleration data are shown from the fifth to the eighth row of Table 4. The RMS value, vibration energy concentration coefficient, and the scores of each sensor are shown in rows 6, 7, and 8, respectively. Furthermore, the aggregate score of the wind field data module is $5 > A = 3$ and that of the vibration acceleration data module is $4 = B = 4$. Therefore, according to the VIV prediction model, it is determined that VIV occurred on the bridge during this period. Then, the management department can take countermeasures according to Table 1.

6. Conclusion and prospects

The wind field and vibration acceleration data measured from 2014 to 2016 by the anemometers and acceleration sensors installed on the bridge were analyzed in detail in this study. The VIV prediction model was established based on the results of big data analysis. The main conclusions are as follows:

- 1) Four parameters, including the average wind speed, wind direction, RMS value, and vibration energy concentration coefficient, were determined as indexes used to establish the prediction model of VIV for the Zhoushan Sea Crossing Bridge.
- 2) The monitoring data of each sensor is less likely to simultaneously satisfy all parameters of the wind field and vibration acceleration data during VIV. Only some of the sensors may meet the threshold values of some parameters when VIV occurs.
- 3) By optimizing the discriminant mechanism of the VIV prediction model, the monitoring data of 2017 was taken as input to test the model. The results showed that the recognition rate of VIV was 89% and the prediction model can be used to warn about the occurrence of VIV in practical engineering.

Moreover, the indexes of different sensors have the same weight coefficient value in the prediction model, which causes some VIV events to remain unrecognized. More VIV monitoring data are needed to optimize the weight coefficient of each sensor to optimize the prediction model and improve the VIV recognition rate.

Disclosure statement

No potential conflict of interest was reported by the authors.

Acknowledgement

The authors would like to thank the National Natural Science Foundation of China, China [grant numbers 51778472, 51878493, 51678437] and Zhejiang Provincial Basic Public Welfare Research Project of China, Zhejiang Province [grant number LGF18E080002].

References

- Biety, J., Delaunay, D., Conti, E., 1995. Comparison of full-scale measurement and computation of wind effects on a cable-stayed bridge. *J. Wind Eng. Ind. Aerodyn.* 57, 225–235.

- Cantero, D., Oiseth, O., Ronnquist, A., 2018. Indirect monitoring of vortex induced vibration of suspension bridge hangers. *Struct. Health Monit.* 17 (4), 837–849.
- Chen, Zhengqing, Huang, Zhiwen, 2015. Analysis of main factors influencing allowable magnitude of vertical vortex-induced vibration of long-span bridge. *China J. Highw. Transp.* 28 (9), 30–37.
- Ehsan, F., Scanlan, R.H., 1990. VIVs of flexible bridges. *J. Eng. Mech.* 116 (6), 1392–1441.
- Frandsen, J.B., 2001. Simultaneous pressures and accelerations measured full-scale on the Great Belt East suspension bridge. *J. Wind Eng. Ind. Aerodyn.* 89, 95–129.
- Hua, X.G., Chen, Z.Q., Chen, W., Niu, H.W., Huang, Z.W., 2015. Investigation on the effect of vibration frequency on VIVs by section model tests. *Wind Struct.* 20 (2), 349–361.
- Kubo, Y., 2004. Prospects for the suppression of aerodynamic vibrations of a long-span bridge using boundary-layer control. *J. Vib. Control* 10 (9), 1359–1373.
- Laima, S.J., Li, H., Chen, W.L., Li, F.C., 2013. Investigation and control of vortex induced vibration of twin box girders. *J. Fluids Struct.* 39, 205–221.
- Li, H., Laima, S.J., Ou, J.P., Zhao, X.F., Zhou, W.S., Yu, Y., Li, N., Liu, Z.Q., 2011. Investigation of vortex-induced vibration of a suspension bridge with two separated steel box girders based on field measurements. *Eng. Struct.* 33 (6), 1894–1907.
- Li, Shanwu, Laima, Shujin, Hui, Li, 2017. Cluster analysis of winds and wind-induced vibrations on a long-span bridge based on long-term field monitoring data. *Eng. Struct.* 138, 245–259.
- Ma, Rujin, Xu, Shiqiao, Wang, Dalei, Chen, Airong, 2018. Vehicle models for fatigue loading on steel box-girder bridges based on weigh-in-motion data. *Struct. Infrastruct. Eng.* 14 (6), 701–713.
- MacDonald, J.H.G., Irwin, P.A., Fletcher, M.S., 2002. VIVs of the Second Severn Crossing cable-stayed bridge - full-scale and wind tunnel measurements. *Struct. Build.* 152 (2), 123–134.
- Marra, A.M., Mannini, C., Bartoli, G., 2015. Measurements and improved model of vortex induced vibration for an elongated rectangular cylinder. *J. Wind Eng. Ind. Aerodyn.* 147, 358–367.
- Marra, A.M., Mannini, C., Bartoli, G., 2017. Wind tunnel modeling for the vortex induced vibrations of a Yawed bridge tower. *J. Bridge Eng.* 22 (5), 1706–1720.
- Mashnad, M., Jones, N.P., 2014. A model for vortex induced vibration analysis of long-span bridges. *J. Wind Eng. Ind. Aerodyn.* 134, 96–108.
- Yang, Y.X., Ma, T.T., Ge, Y.J., 2015. Evaluation on bridge dynamic properties and vortex induced vibration performance based on wind tunnel test and field measurement. *Wind Struct.* 20 (6), 719–737.
- Yoshimura, T., Mizuta, Y., 2001. Prediction of vortex-induced oscillations of a bridge girder with span wise varying geometry. *J. Wind Eng. Ind. Aerodyn.* 89, 1717–1728.
- Yu, Chuanjin, Li, Yongle, Zhang, Minjin, Zhang, Yi, Zhai, Guanghao, 2019. Wind characteristics along bridge catwalk in a deep-cutting gorge from field measurements. *J. Wind Eng. Ind. Aerodyn.* 186, 94–104.
- Zhou, Z.Y., Yang, T., Ding, Q.S., Ge, Y.J., 2015. Mechanism on suppression in vortex induced vibration of bridge deck with long projecting slab with countermeasures. *Wind Struct.* 20 (5), 643–660.
- Zuo, D., Jones, N.P., Main, J.A., 2008. Field observation of vortex-and rain-wind-induced stay-cable vibrations in a three-dimensional environment. *J. Wind Eng. Ind. Aerodyn.* 96, 1124–1133.